Introduction to Markov Chains

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A stochastic process

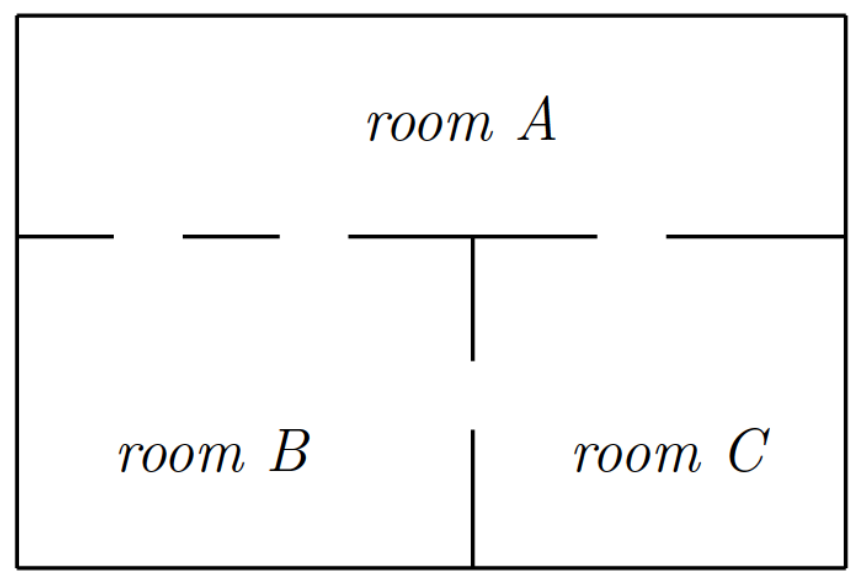
- A *stochastic process* means a system for which there are observations at certain times $t = 0, 1, 2, \ldots$
- The outcome, that is, the observed value at each time is a random variable $\{X_t\} = \{X_0, X_1, X_2, \ldots\}$.
- This means that, for each observation at a certain time, there is a certain probability to get a certain outcome.
- In general, that probability depends on what has been obtained in the previous observations.
- The more observations we have made, the better we can predict the outcome at a later time.

A Markov process

- A *Markov process* is a process where all information that is used for predictions about the outcome at some time is given by the latest observation(s).
- Its result and the time lapsed since then are everything we need for assigning a probability for a new observation.
- Whatever is observed before that latest observation(s) has no influence on the outcome we next want to attain.

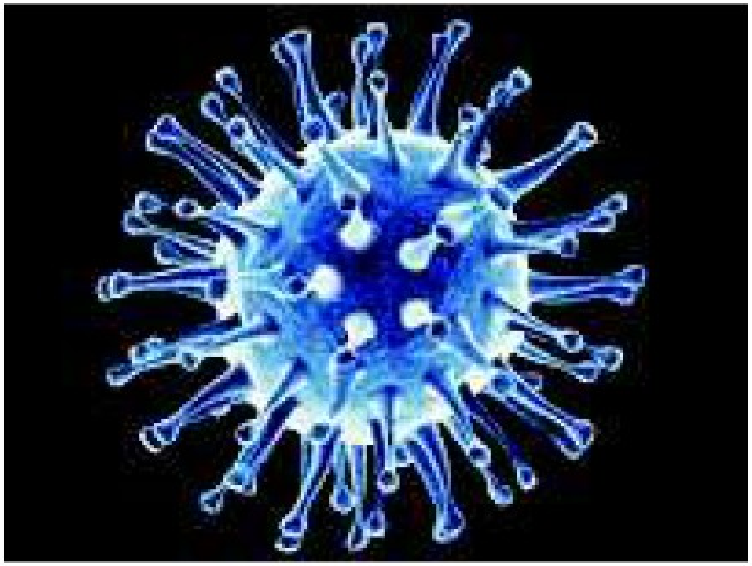
Application 1: Trained mouse

- A trained mouse lives in the house shown.
- A bell rings at regular intervals, and the mouse is trained to change rooms each time it rings.
- When it changes rooms, it is equally likely to pass through any of the doors in the room it is in.
- Approximately what fraction of its life will it spend in each room?



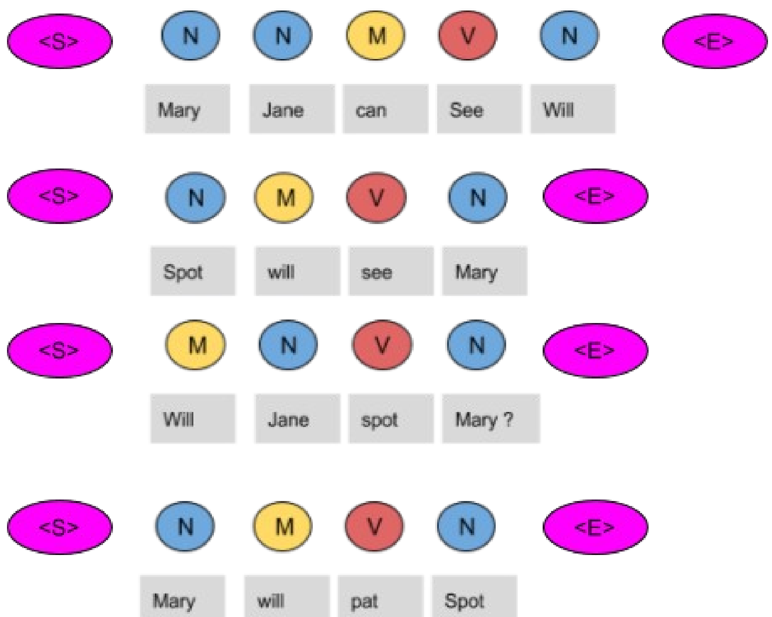
Application 2: Mutating virus

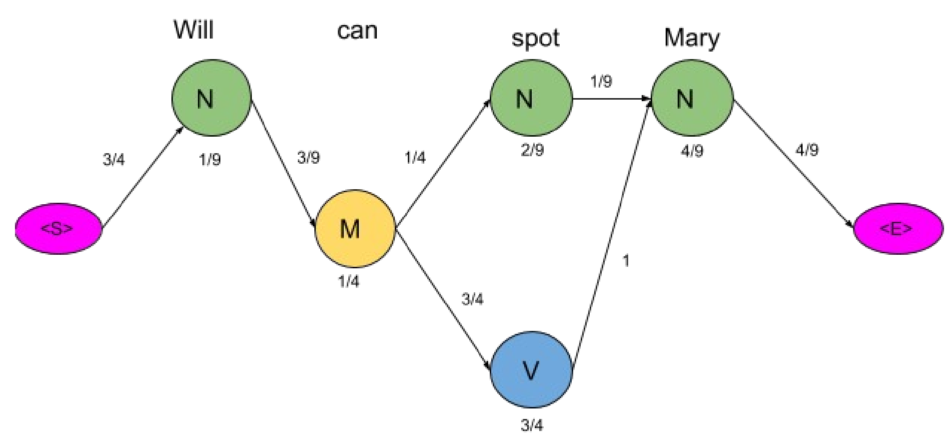
- A virus can exist in N different strains.
- At each generation the virus mutates with probability $\alpha \in (0,1)$ to another strain which is chosen at random.
- What is the probability that the strain in the n-th generation of the virus is the same as that in the 0-th?



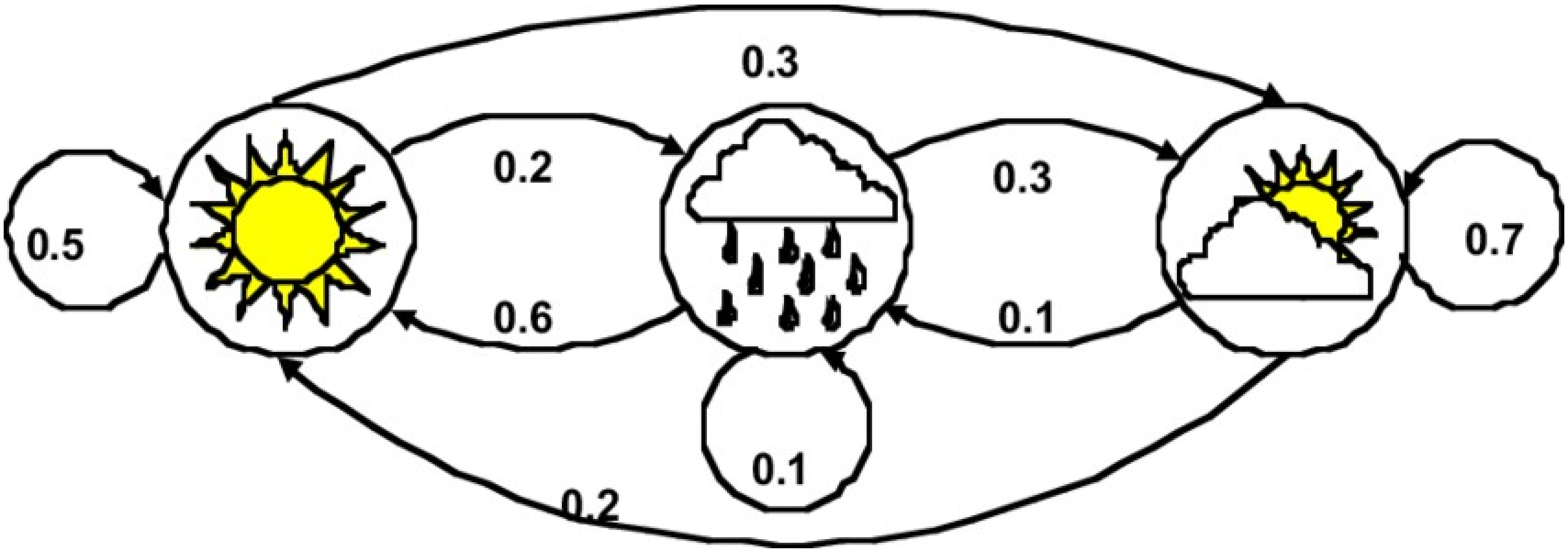
Application 3: PoS Tagging

Data:
Mary Jane can see Will.
Spot will see Mary.
Will Jane spot Mary?
Mary will pat Spot

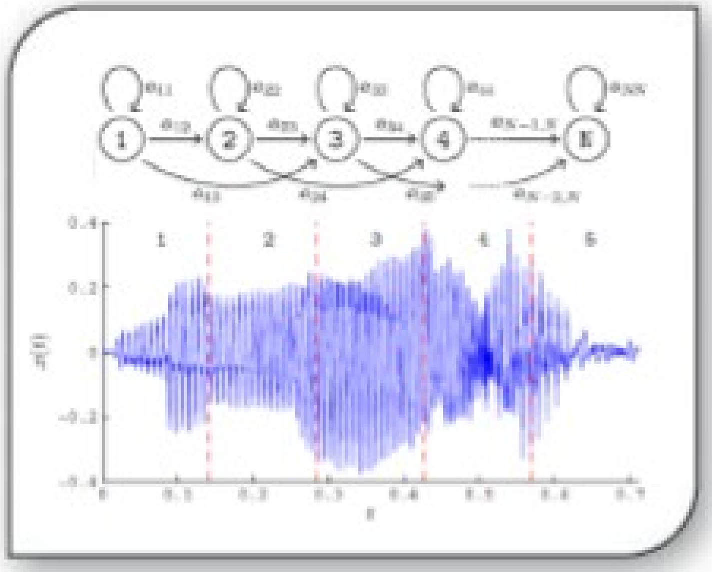


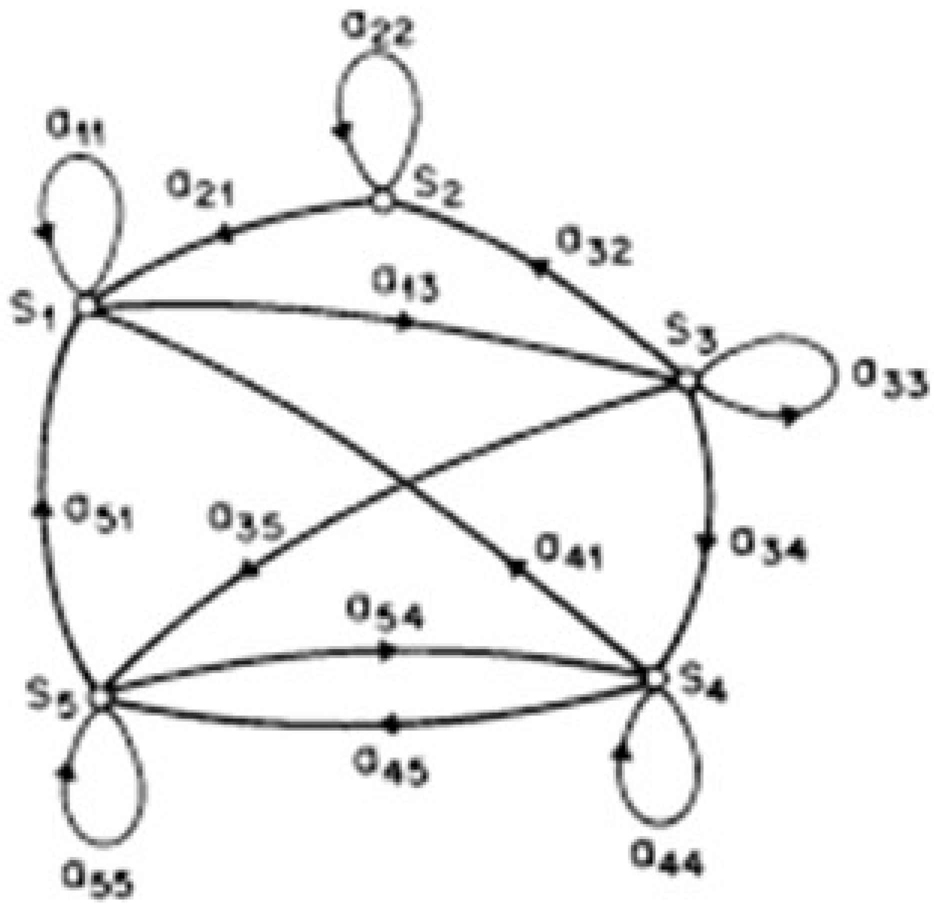


Application 4: Weather forecast

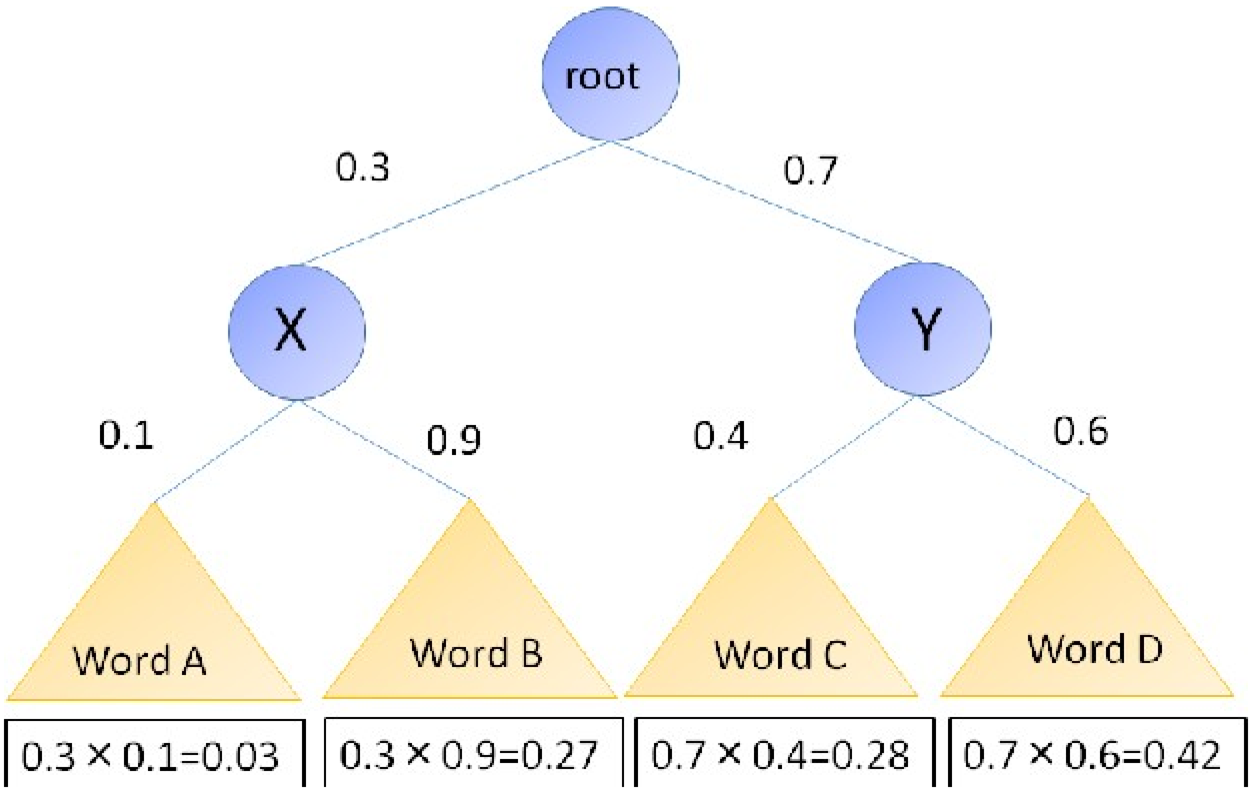


Application 5: Speech recognition

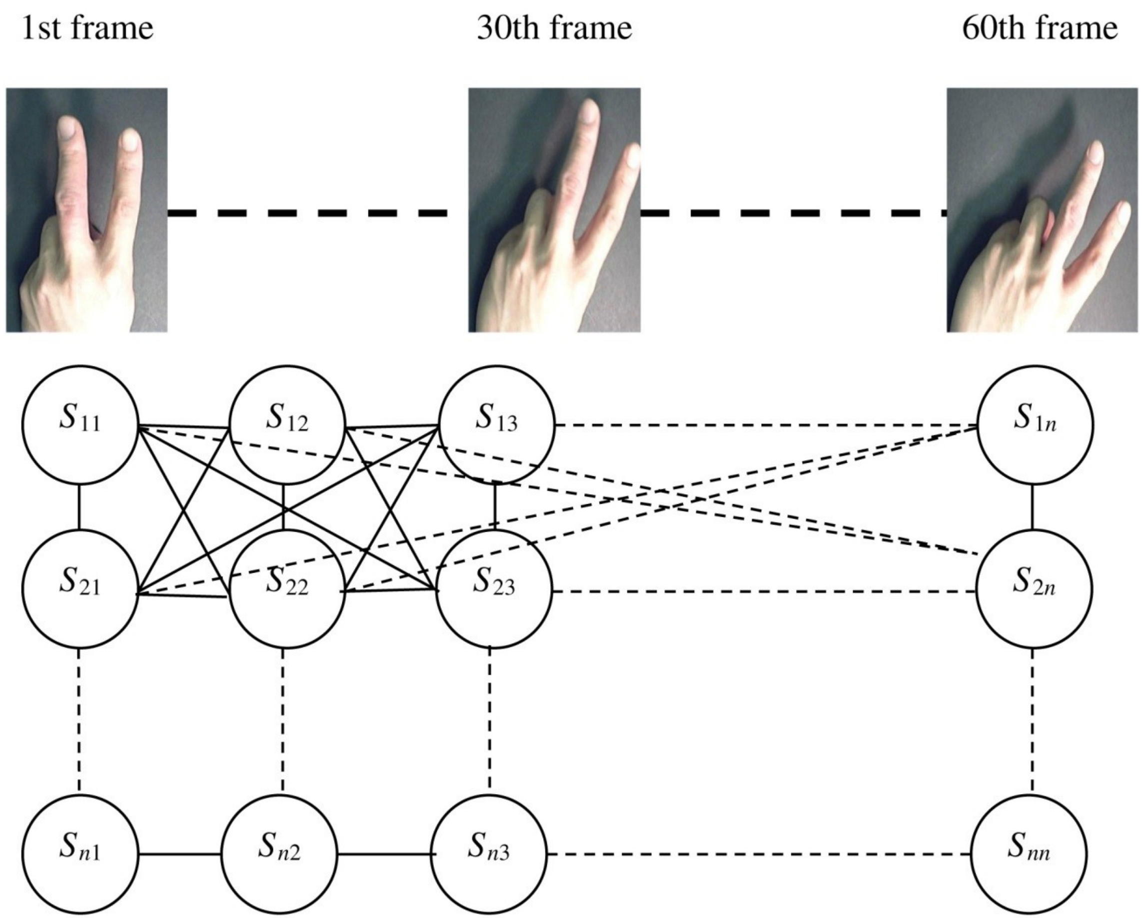




Application 6: Surrounding Word Sense Disambiguation



Application 7: Hand gesture recognition



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